

■ AI/ML Internship – Day 79 Notes & Tasks

● Module 9: RAG Evaluation – Measuring Accuracy, Precision, Recall, Hallucinations & Faithfulness

📌 Part 1: Recap of Day 78

Yesterday we learned:

- ☒ Conversation Memory
- ☒ Chat History
- ☒ Buffer Memory
- ☒ Window Memory
- ☒ Summary Memory
- ☒ Memory in RAG

Today we will learn:

- What is RAG Evaluation?
 - Why Evaluation is Important
 - Precision
 - Recall
 - Faithfulness
 - Hallucination
 - Context Relevance
 - Answer Relevance
 - Evaluation Frameworks
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📌 Part 2: What is RAG Evaluation?

Definition

RAG Evaluation is the process of measuring how accurately a Retrieval-Augmented Generation (RAG) system retrieves relevant information and generates correct responses.

Simple Definition

RAG Evaluation helps answer questions like:

- Is the chatbot giving correct answers?
 - Is it retrieving the right documents?
 - Is it making up information?
 - Can users trust its responses?
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Part 3: Why Do We Need Evaluation?

Imagine two chatbots answering the same question.

Chatbot A

Internship duration is 3 months.

(Source: Internship.pdf Page 5)

Chatbot B

Internship duration is 6 months.

(No source provided.)

Which chatbot is more trustworthy?

 Chatbot A

Evaluation helps identify which system performs better.

Part 4: RAG Evaluation Workflow

User Question

|



Retriever

|



Retrieved Documents

|



LLM

|



Generated Answer

|



Evaluation Metrics

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Performance Score

Part 5: Precision

Definition

Precision measures how many of the retrieved documents are actually relevant.

Formula

$$\text{Precision} = \text{Relevant Retrieved Documents} / \text{Total Retrieved Documents}$$

Example

Retriever returns:

- HR Policy 

- Leave Policy 
- Sports Event 
- Finance Report 

Only 2 out of 4 documents are relevant.

Precision:

$$2 / 4 = 0.5 \text{ (50\%)}$$

High Precision

Most retrieved documents are useful.

Part 6: Recall

Definition

Recall measures how many relevant documents were successfully retrieved.

Formula

$$\text{Recall} = \text{Relevant Retrieved Documents} / \text{Total Relevant Documents}$$

Example

Database contains:

- HR Policy
- Leave Policy
- Attendance Policy
- Holiday List

Retriever finds:

- HR Policy
- Leave Policy

Recall:

$$2 / 4 = 50\%$$

High Recall means fewer relevant documents are missed.

✦ Part 7: Precision vs Recall

Precision

Recall

Focuses on correctness

Focuses on completeness

Fewer irrelevant results

Fewer missing results

High precision = clean results High recall = more complete search

✦ Part 8: Hallucination

Definition

A Hallucination occurs when the LLM generates information that is not supported by the retrieved documents.

Example

Document:

Internship duration is 3 months.

AI Response:

Internship duration is 6 months and includes an overseas internship.

✗ The extra information is not present in the document.

This is hallucination.

✦ Part 9: Faithfulness

Definition

Faithfulness measures whether the AI's answer is supported by the retrieved context.

Example

Retrieved Context:

Training begins on August 1.

Answer:

Training begins on August 1.

✅ Faithful Answer

Answer:

Training begins on September 1.

❌ Not Faithful

📌 Part 10: Context Relevance

Definition

Measures whether the retriever selected the correct document chunks.

Example:

User:

Explain leave policy.

Retriever returns:

- Leave Policy ✅
- HR Rules ✅

Good Context.

Retriever returns:

- Sports Event ❌
- Company Picnic ❌

Poor Context.

📌 Part 11: Answer Relevance

Measures whether the generated answer actually answers the user's question.

Example:

Question:

What is Machine Learning?

Answer:

Machine Learning is a branch of AI that enables computers to learn from data.

✅ Relevant Answer

Question:

What is Machine Learning?

Answer:

Python is a programming language.

❌ Not Relevant

📌 Part 12: Common RAG Evaluation Metrics

Metric	Measures
Precision	Correct retrieved documents
Recall	Retrieved all relevant documents
Hallucination	False information generated
Faithfulness	Answer supported by context
Context Relevance	Quality of retrieved chunks
Answer Relevance	Whether the answer addresses the question

📌 Part 13: Evaluation Tools

Popular frameworks include:

- Ragas
- DeepEval
- LangSmith
- TruLens
- Phoenix (Arize AI)

These tools help developers automatically evaluate RAG systems.

Part 14: Best Practices

- ✓ Use trusted documents.
 - ✓ Retrieve high-quality chunks.
 - ✓ Display source citations.
 - ✓ Test with many different questions.
 - ✓ Continuously improve retrieval and prompts.
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Part 15: Advantages of Evaluation

- Improves chatbot accuracy.
 - Detects hallucinations.
 - Builds user trust.
 - Identifies retrieval problems.
 - Supports continuous improvement.
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Part 16: Limitations

- Requires good test datasets.
- Some metrics require manual review.
- Different evaluation tools may produce different scores.

- Evaluation should be continuous as documents and models change.
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Part 17: Theory Questions

1. What is RAG Evaluation?

RAG Evaluation is the process of measuring how well a RAG system retrieves relevant information and generates accurate answers.

2. What is Precision?

Precision measures the percentage of retrieved documents that are actually relevant.

3. What is Recall?

Recall measures how many of the relevant documents were successfully retrieved.

4. What is Hallucination?

Hallucination occurs when an AI generates information that is not supported by the retrieved documents.

5. What is Faithfulness?

Faithfulness measures whether the generated answer is fully supported by the retrieved context.

Part 18: Interview Questions

Why is RAG Evaluation important?

It ensures that the chatbot retrieves relevant information, generates accurate answers, and minimizes hallucinations.

What is the difference between Precision and Recall?

Precision focuses on the quality of retrieved documents, while Recall focuses on retrieving all relevant documents.

How can hallucinations be reduced?

By improving document retrieval, using strong prompt engineering, displaying source citations, and restricting the LLM to the retrieved context.

What is Context Relevance?

It measures whether the retrieved document chunks are appropriate for answering the user's question.

Name three RAG evaluation frameworks.

- Ragas
 - LangSmith
 - DeepEval
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Tasks for Day 79

Theory

1. Define RAG Evaluation.
 2. Differentiate Precision and Recall.
 3. Explain Hallucination with an example.
 4. What is Faithfulness?
 5. Why is Context Relevance important?
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Practical Task 1

Ask 10 different questions to your RAG chatbot.

For each question:

- Check whether the retrieved documents are relevant.
 - Check whether the final answer is correct.
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Practical Task 2

Create a table with the following columns:

| Question | Retrieved Documents | Correct Answer? | Hallucination? | Faithful? |

Fill the table for at least 10 test questions.

Practical Task 3

Intentionally ask a question that is not present in the uploaded documents.

Observe whether:

- The chatbot says "I don't know."
 - Or whether it hallucinates.
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Practical Task 4

Compare retrieval using:

- Top-1 result
- Top-3 results
- Top-5 results

Write your observations on answer quality.

Practical Task 5

Research one RAG evaluation framework (such as Ragas or DeepEval) and summarize:

- Features
- Advantages
- Use cases

